



The Living Map: Spatial Memory for Emergency Robots

IEEE RAS × IEEE AES – Tunisia Section Chapters

TSYPI4 TECHNICAL CHALLENGE

SCOPE & TOPIC

In disaster response, underground exploration and hazardous inspections, autonomous robots may operate where GPS and communication networks are unavailable.

The challenge focuses on resilient communication in disconnected environments, allowing a first robot to leave critical information through small radio beacons so that a second robot can retrieve it and continue the mission without starting from zero.

Challenge problem

In collapsed buildings, underground mines, fire-damaged structures or contaminated zones, robots must explore, sense, detect hazards, locate victims and communicate despite limited connectivity.

When a robot fails, leaves or loses power, the knowledge it collected may disappear with it.

The challenge is to build a resilient autonomous system that can:

Sense → Communicate → Preserve Critical Information
→ Enable Mission Continuity

Challenge goal

Design and prototype a two-robot system that gives an unmapped space its own memory.

Choose one environment:

- Urban Search & Rescue
- Fire / Hazardous Building
- Mines / Tunnels
- Nuclear / Contaminated Zone
- The system must:
- Build a Writer Robot that autonomously explores a GPS-denied space.
- Detect at least 2 event types.
- Deploy small radio beacons where relevant.

- Design beacon messages containing what was found, where to go and when it was written.
- Build an Outside Network Area connecting the disconnected zone to the outside world.
- Translate robot coordinates into real-world GPS coordinates.
- Transfer information to a distant command post.
- Brief the Executor Robot before entry.
- Build an Executor Robot that receives a mission and navigates using inherited beacon information.
- Display a live map at the command post.

Challenge instructor

Each team must:

- Complete registration before the deadline.
- Choose one environment.
- Detect at least 2 event types.
- Use at least 2 physical robots:
 - Writer
 - Executor
- Implement a functional Outside Network Area.
- Ensure all communication between the inside and outside passes through the Outside Network Area.
- No direct link is allowed between the robots and the command post.

System chain



Technical focus

Writer Robot

- Autonomous navigation
- Exploration
- Decision-making in GPS-denied environments

Event Detection & Beacon Deposition

- Sensing
- Deciding what information to record
- Beacon drop mechanism

Beacon Design

- Compact message
- RF broadcast
- Message aging mechanism

Frame Translation

Private robot coordinates → Real-world GPS coordinates

Outside Network Area

- Command-post connection
- Satellite / wireless communication
- Data reception and forwarding

Executor Robot

- Mission briefing
- Beacon-guided navigation
- Mission execution

Technical Documentation

- System architecture
- Data flow
- Additional technical diagrams

Ressources

[System Architecture Diagram](#)

Phase 1: Initial phase

Required Deliverables

- GitHub Repository
- Simulation Demo
- Short Technical Report: max. 6 pages
- Implementation Plan
- Failure Cases
- Technical Solution & Architecture

The solution must address:

- Writer robot autonomy
- Event detection & beacon deposition
- Beacon message & signal design
- Frame translation
- Outside network architecture
- Executor robot
- System architecture & data flow

Phase 2: Final phase

Finalists must provide:

- Pitching & Q&A
- Complete physical prototype
- Proposed solution
- Technologies used
- Project architecture
- User manual
- GitHub Repository

Pitching

English

5min pitching

2min Q&A

Scoring

TOTAL SCORE: 100 POINTS

Initial Phase: 45 pts

- GitHub Repository: 4 pts
- Simulation Demo: 8 pts
- Short Technical Report: 2 pts
- Innovation, Originality & Creativity: 5 pts
- Implementation Plan: 3 pts
- Failure Cases: 3 pts

Technical aspects of the solution

- Writer Robot autonomy: 5 pts
- Event detection & beacon deposition: 5 pts
- Beacon message & signal design: 5 pts
- Frame translation: 5 pts
- Outside network area: 5 pts
- Executor Robot: 5 pts
- Technical solution diagrams: 5 pts

Final Phase: 50 pts

- Pitching and Q&A: 8 pts
- Quality of proposed solutions: 8 pts
- Technologies used: 5 pts
- Physical prototype: 15 pts
- Project Architecture: 5 pts
- User Manual: 4 pts
- GitHub Repository: 5 pts

BONUS: 5 pts

- At least 2 IEEE AESS members in the team: 2 pts
- At least 2 RAS members in the team: 2 pts
- At least 1 YP member with both RAS & AESS memberships: 1 pt

Important dates

- Challenge Infoession: **14/09/2026**
- Phase 1 Submission Deadline: **05/10/2026**
- Final Submission Deadline: **01/12/2026**

Important dates

2 WINNING TEAMS

- **1st Place:** \$100 USD + Certificate
- **2nd Place:** \$50 USD + Certificate

Submission

Submit your project and show us what you've built.

Contact



Please contact us through the following email address aess@ieee.tn or ras@ieee.tn for any inquires or clarifications.